

Efficiency of Ferroelectric Field-Effect Transistors: An Experimental Study

Nujhat Tasneem¹, Graduate Student Member, IEEE, Muhammad M. Islam, Student Member, IEEE, Zheng Wang², Graduate Student Member, IEEE, Zijian Zhao, Graduate Student Member, IEEE, Navnidhi Upadhyay, Sarah F. Lombardo, Hang Chen, Jae Hur³, Member, IEEE, Dina Triyoso, Steven Consiglio⁴, Senior Member, IEEE, Kanda Tapily⁵, Senior Member, IEEE, Robert Clark⁶, Member, IEEE, Gert Leusink, Santosh Kurinec⁷, Life Fellow, IEEE, Suman Datta⁸, Fellow, IEEE, Shimeng Yu⁹, Senior Member, IEEE, Kai Ni¹⁰, Member, IEEE, Matthias Passlack¹¹, Fellow, IEEE, Winston Chern, Member, IEEE, and Asif Khan¹², Member, IEEE

Abstract—While the theoretical maximum of the memory window ΔV_t in a ferroelectric field-effect transistor (FEFET) is $2E_C t_F$, E_C and t_F being coercive field and FE thickness, respectively, experimentally ΔV_t is observed to be much less than that, even in the case of complete polarization switching in the ferroelectric layer. This occurs because trapped charges in the gate oxide stack partially or completely screen the ferroelectric polarization, only a fraction of which gets “electrostatically” reflected in the semiconductor channel. In this article, we provide a generalized experimental framework to quantify the efficiency of practical FEFETs in converting the switched ferroelectric polarization into the memory window. To that end, we propose three efficiency metrics: 1) switched ferroelectric polarization ΔP_F to memory window conversion ratio, $\eta_{MW} = \Delta V_t / \Delta P_F$; 2) switched ferroelectric polarization to the semiconductor charge (ΔQ_S) conversion efficiency or the charge conversion efficiency $\eta_C = \Delta Q_S / \Delta P_F$; and 3) the ratio of memory window to the width of polarization versus gate voltage

hysteresis characteristics (ΔV_p) or the voltage conversation efficiency $\eta_V = \Delta V_t / \Delta V_p$. We measure and compare these efficiency metrics in n-type and p-type FEFETs, fabricated in different facilities by different groups. In all the cases, we find that the maximum values of memory window efficiency $(\eta_{MW})_{\max}$, charge conversion efficiency $(\eta_C)_{\max}$, and voltage conversion efficiency $(\eta_V)_{\max}$ are in the range: 2.5–5.5 Vm²/C, 4%–10%, and 5%–20%, respectively.

Index Terms—Efficiency, ferroelectric field-effect transistor (FEFET), memory window.

I. INTRODUCTION

FERROELECTRIC field-effect transistors (FEFETs) have emerged as an attractive non-volatile memory element for embedded, data-centric applications, thanks to the discovery of ferroelectricity in complementary metal–oxide–semiconductor (CMOS)-compatible binary oxides based on hafnia and zirconia [1]. These fluorite-structure ferroelectric oxides are energy- and area-efficient, suitable for merged logic–memory functionalities, and compatible with high-volume semiconductor manufacturing processes [2]–[9]. One of the key design objectives of an FEFET is to maximize the memory window ΔV_t for a given write voltage. The theoretical maximum of ΔV_t is $2E_C t_F$, E_C , and t_F being coercive field and FE thickness, respectively [10]. However, all reports till date have shown that $\Delta V_t \ll 2E_C t_F$, even when the ferroelectric layer in an FEFET operates on a major loop in the polarization (P_F) versus gate voltage (V_G) characteristics (i.e., the ferroelectric layer switches completely). Recent reports have shown that this occurs due to the screening of ferroelectric polarization at the interfaces by traps and that only a fraction of the ferroelectric polarization gets electrostatically reflected in the semiconductor channel, thereby reducing the memory window significantly below its theoretical limit [11]–[14].

Fig. 1 illustrates the discrepancy between the theoretical framework and the experimental observations with regard to the memory window in an FEFET. The experimental setups for measuring the polarization versus gate voltage (P – V_G) and drain current versus gate voltage (I_D – V_G) characteristics

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Nujhat Tasneem, Muhammad M. Islam, Zheng Wang, Sarah F. Lombardo, Hang Chen, Jae Hur, Shimeng Yu, Winston Chern, and Asif Khan are with the School of Electrical and Computer Engineering (ECE), Georgia Institute of Technology, Atlanta, GA 30318 USA (e-mail: nujhat.tasneem@gatech.edu).

Zijian Zhao, Santosh Kurinec, and Kai Ni are with the Department of Electrical and Microelectronic Engineering, Rochester Institute of Technology, Rochester, NY 14623 USA.

Navnidhi Upadhyay and Suman Datta are with the Department of Electrical Engineering, University of Notre Dame, Notre Dame, IN 46556 USA.

Dina Triyoso, Steven Consiglio, Kanda Tapily, Robert Clark, and Gert Leusink are with TEL Technology Center, America, LLC, Albany, NY 12203 USA.

Matthias Passlack is with Corporate Research, Taiwan Semiconductor Manufacturing Company Ltd., San Jose, CA 95134 USA.

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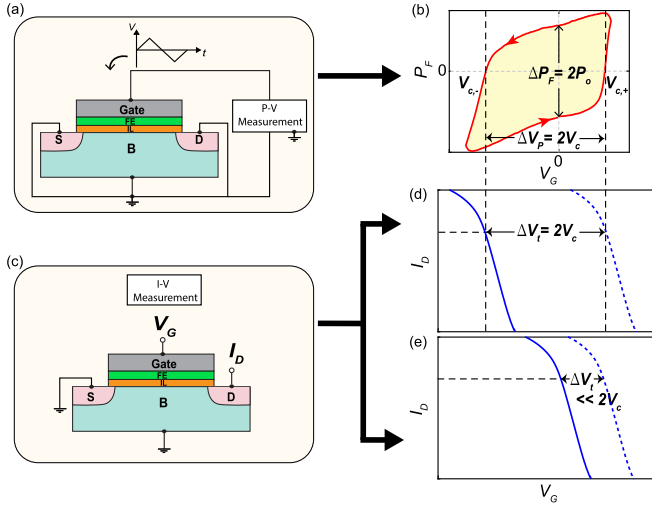


Fig. 1. (a) Experimental setup to obtain the ferroelectric polarization versus gate voltage (P_F - V_G) characteristics between gate (G) and shorted source (S), drain (D), and body (B) terminals of the FEFET. (b) Memory window, MW, acquired from the P_F - V_G loop is denoted by ΔV_P . (c) Experimental setup to obtain the I_D - V_G characteristics of the FEFET. (d) Theoretically obtained maximum MW ($\Delta V_t \approx 2V_C = 2E_{CTF}$) from the I_D - V_G curve, while (e) shows the experimentally extracted MW ($\Delta V_t \ll 2V_C$).

of an FEFET are shown in Fig. 1(a) and (c), respectively. If a given combination of the program and erase voltages (i.e., the sweep range of V_G) leads to a complete polarization switching, the window of the P_F versus V_G hysteresis loop, ΔV_P , will follow the relationship $\Delta V_P = V_{C,+} - V_{C,-} = 2V_C = 2E_{CTF}$, where $V_{C,+}$ and $V_{C,-}$ are the coercive voltages of the ferroelectric layer only in the positive and negative voltage ranges, respectively, and V_C is the average of $V_{C,+}$ and $V_{C,-}$ [i.e., $V_C = (V_{C,+} - V_{C,-})/2$] [see Fig. 1(b)]. If the sweep range of V_G leads to a partial polarization switching, $\Delta V_P < 2E_{CTF}$. Ideally, in either case: a complete polarization switching or partial, if this polarization is fully translated to the semiconductor as charge carriers, $\Delta V_t \approx \Delta V_P$ [see Fig. 1(d)]. However, experimentally, interfacial polarization screening by trapped charges makes the electrostatic coupling between the ferroelectric layer and the semiconductor channel weak; hence, ΔV_t is much lower than its theoretical counterpart—i.e., $\Delta V_t \ll \Delta V_P$ [see Fig. 1(e)].

Based on this perspective, an important metric for an FEFET would be one that measures how efficient the switched ferroelectric polarization is in affecting a shift in the threshold voltage V_t . We propose an efficiency metric η_{MW} , which we refer to as the (switched) polarization to memory window conversion ratio, defined as follows:

$$\eta_{MW} = \frac{\Delta V_t}{\Delta P_F}. \quad (1)$$

In the ideal case, the maximum value of η_{MW} , $(\eta_{MW})_{\max} = 2V_C/2P_o = 2E_{CTF}/2P_o$, where P_o is the remnant polarization of the ferroelectric layer. In a more general case that includes a partial polarization switching, $(\eta_{MW})_{\max} = \Delta V_P/2P_o$.

At a given V_G at which the device is in strong inversion, the difference between the inversion carrier density between the

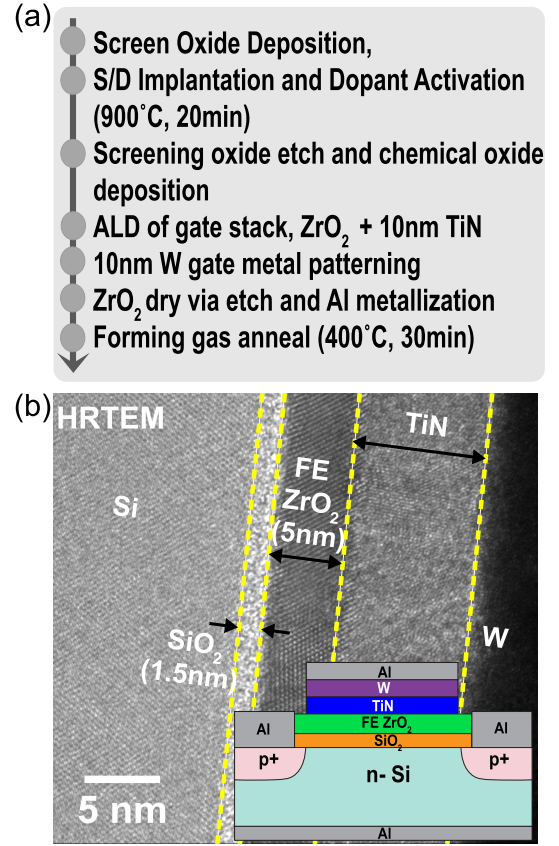


Fig. 2. (a) Fabrication process flow of the FEFET. (b) HR-TEM image of the FEFET gate-stack showing clear interfaces, with the inset showing the schematic of the FEFET.

programmed and erased states is given by $\Delta Q_s = C_{ON} \Delta V_t$, where C_{ON} is the ON-state capacitance of the FEFET. As such, another metric of FEFET performance can be defined, which we refer to as ferroelectric polarization to semiconductor charge conversion efficiency, as follows:

$$\eta_c = \frac{\Delta Q_s}{\Delta P_F} = C_{ON} \eta_{MW}. \quad (2)$$

The ratio of the memory window to its theoretical maximum is another useful metric, defined as follows:

$$\eta_v = \frac{\eta_{MW}}{\eta_{MW_{\max}}} = \frac{\Delta V_t}{\Delta V_P}. \quad (3)$$

We refer to this metric as the voltage conversion efficiency.

In this study, we establish an experimental protocol to quantify the polarization to MW conversion ratio and the charge and voltage conversion efficiencies in a ferroelectric field-effect transistor (FEFET). The protocol consists of a sequence of three measurements: 1) the characterization of the memory window ΔV_t in the I_D - V_G characteristics using a write-read-write-read pulse sequence; 2) the characterization of the switched polarization ΔP_F using positive-up-negative-down (PUND) method; and 3) the characterization of the switched polarization ΔP_F and memory window ΔV_P in the P_F versus V_G characteristics for slow sweeps of V_G . We first carry out the protocol on a p-type ZrO_2 FEFET and quantify the polarization to MW conversion ratio and the

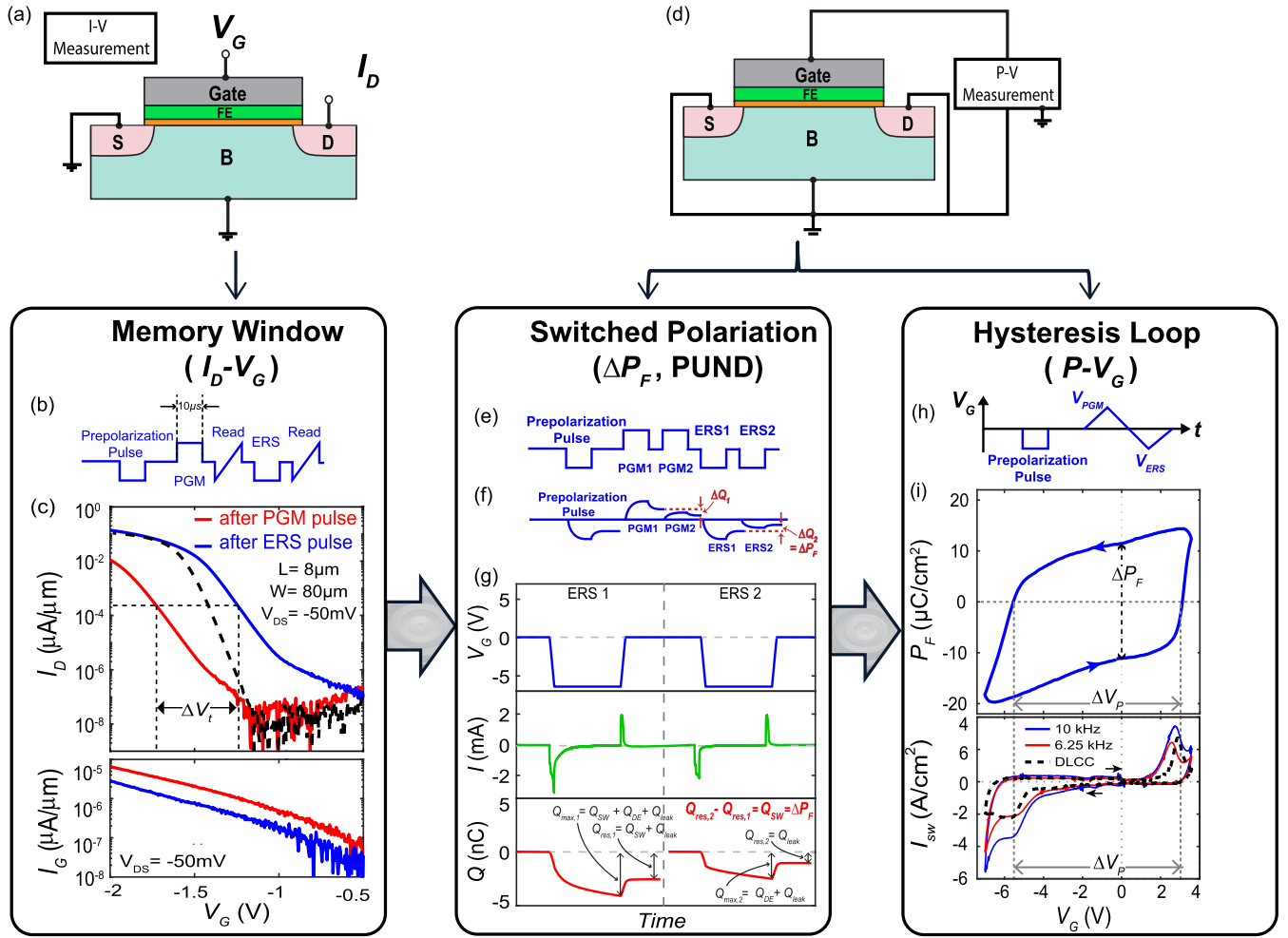


Fig. 3. Measurement protocol of FEFET metrics—memory window (I_D – V_G): (a) experimental setup and (b) pulse sequence to determine ΔV_t of the FEFET. (c) FEFET I_D – V_G and I_G – V_G characteristics with extracted $\Delta V_t \approx 0.5$ V. $I_G \ll I_D$. The subthreshold slope (SS) of the I_D – V_G curve before switching (black dotted curve) is ~ 70 mV/dec. However, SS becomes ~ 110 mV/dec after both the PGM (red curve) and ERS pulses (blue curve) are applied on the device. (d) Next two experiments were conducted by using the experimental setup shown for the P–V measurement, where the pulses were applied to the gate, and the ground was connected to the shorted S, D, and B terminals. Switched polarization (ΔP_F , PUND): (e) pulse sequence to determine ΔP_F and (f) corresponding charge plot. The difference between the residual charges ($Q_{res,1}$ and $Q_{res,2}$) in this two-pulse scheme gives the switching charge across the ferroelectric gate capacitor ΔP_F as detailed in (g). Hysteresis loops (P – V_G): (h) triangular pulses to determine (i) P_F – V_G hysteresis loops obtained between G and shorted S, D, and B terminals. ΔP_F and ΔV_P are obtained from the hysteresis loop as indicated in (i). I_{sw} – V_G curves obtained for 6.25 kHz (red), 10 kHz (blue), and after performing DLCC (black dotted) are also shown in the second panel of (i).

charge and voltage conversion efficiencies. The universality of the suggested measurement technique is further verified in n-type $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ FEFETs fabricated in different facilities.

II. FEFET FABRICATION

A p-type FEFET with the gate-stack composed of 10 nm of TiN, 5 nm of ZrO_2 , and 1.5 nm of chemical oxide (SiO_2) was fabricated using a simple gate-last process flow shown in Fig. 2(a) and the same devices as described in [14]. The source S and drain D are formed using a 15-keV BF_2 ion implant with a dose of $4 \times 10^{15} \text{ cm}^{-2}$. After activating the dopants, the gate-stack is formed using chemical oxide growth and atomic layer deposition (ALD) of TiN/ ZrO_2 , described in reference [15]. Tungsten is then sputtered, the gate is patterned using RIE, and the ZrO_2 is etched at the vias using BCl_3/Cl_2 RIE. The devices are finalized using a standard interlayer dielectric (ILD) and metallization process followed

by a forming gas anneal (400 °C, 30 min). An n-type HZO FEFET, the results of which are shown later in this manuscript (see Figs. 7 and 8), was fabricated at Georgia Tech using the same process flow as described earlier, except that the source and drain were formed using P ion implant at 20 keV with a dose of 10^{15} cm^{-2} . Fig. 2(b) shows the device schematic and the cross-sectional high-resolution transmission electron microscopy (HR-TEM) image of the FEFET gate-stack showing the layer thicknesses and clear interfaces. The gate length, L_G , and width, W , of the measured device in this article are 8 and 80 μm , respectively.

III. MEASUREMENT FOR EFFICIENCY METRICS

The FEFET characterization protocol consists of three subsequent measurements for a given combination of program and erase voltages (V_{PGM} and V_{ERS} , respectively), as follows.

Step 1 Memory window, ΔV_t : Standard memory characteristics were measured using a pre-polarization pulse—program pulse (PGM)— I_D — V_G read—erase pulse (ERS)— I_D — V_G read sequence as shown in Fig. 3(b). The sweep speed for the I_D — V_G read is 2.5 V/ms. In the case of a p-type FEFET, the negative ERS pulse lowers the absolute value of the threshold voltage ($|V_t|$), while the positive PGM pulse increases it. The memory window, ΔV_t , is extracted from the I_D — V_G curves by taking the difference in threshold voltages after program and erase pulses. In this article, we define V_t as V_G at which $I_D/W = 10^{-4} \mu\text{A}/\mu\text{m}$. Keysight B1500A semiconductor device analyzer was used to obtain the device characteristics. The program and erase pulse widths were 10 μs . Representative I_D — V_G and I_G — V_G curves (I_G being the gate current) for $V_{\text{PGM}} = 3.4 \text{ V}$ and $V_{\text{ERS}} = -6.6 \text{ V}$ are shown in Fig. 3(c). It can also be seen from Fig. 3(c) that the intrinsic sub-threshold swing (SS), i.e., the SS of the I_D — V_G curve before switching (black dotted curve) is much lower, close to the ideal value ($\sim 70 \text{ mV/dec}$), which is indicative of a good quality interfacial oxide (SiO_2). However, the SS after the PGM and ERS voltages is applied, i.e., the SS of the I_D — V_G curve after switching [blue and red curves in Fig. 3(c)] is much higher ($\sim 110 \text{ mV/dec}$) which implies that the charge trapping mostly originates after the ferroelectric layer is switched.

Step 2 Switched polarization, ΔP_F : Immediately after the ΔV_t measurement for a given combination of (V_{PGM} , V_{ERS}), the experimental setup is changed to the one shown in Fig. 3(d) to measure the switched polarization (ΔP_F) using the positive-up–negative-down (PUND) scheme [see Fig. 3(e)] by changing the connection using a switch matrix. PUND pulse sequence is applied at the gate (G) with the shorted source, drain, and body (S, D, and B, respectively) being grounded [see Fig. 3(d)]. Fig. 3(f) shows the corresponding charge plot which is further detailed in Fig. 3(g). After applying a pre-polarization pulse, two subsequent erase (ERS) pulses and two program (PGM) pulses were applied. In both the cases, the first pulse switches the device under test (DUT), while the second pulse only contains the dielectric component and leakages. The difference between the residual charges in this two-pulse scheme gives the switching charge across the ferroelectric gate capacitor, ΔP_F [see Fig. 3(f) and (g)]. The program and erase pulse widths were 10 μs here as well. Fig. 3(g) shows the two subsequent voltage pulses (blue) and the corresponding switching current (green) and details the switched polarization extraction from the corresponding charge plots (red).

Step 3 Hysteresis window ΔV_P and switched polarization, ΔP_F : Finally, the P_F — V_G characteristics is measured by applying a triangular pulsing sequence [see Fig. 3(h)] to G with the same connections (i.e., S,

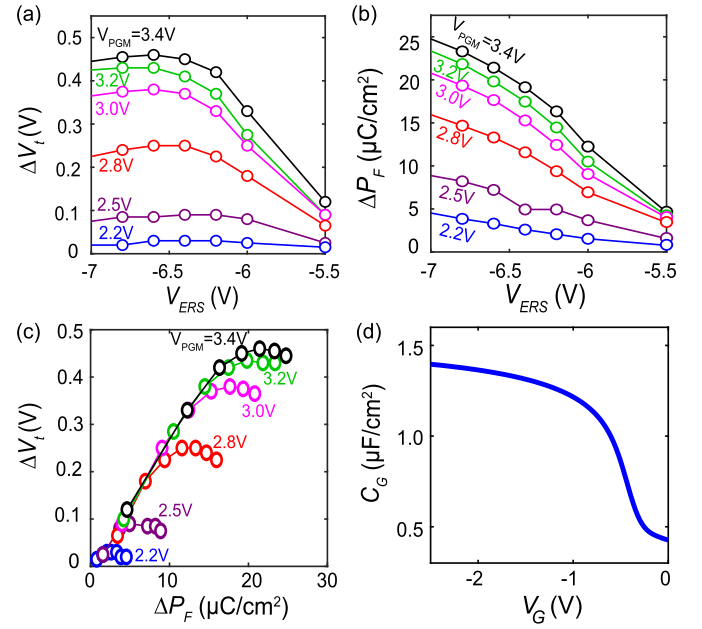


Fig. 4. Variation of (a) ΔV_t and (b) ΔP_F with respect to V_{ERS} , and (c) ΔV_t with respect to ΔP_F for different values of V_{PGM} in ZrO_2 FET. (d) C_G — V_G curve for the p-type ZrO_2 FEFET.

B, and D shorted) at 6.25 and 10 kHz taking into account the dynamic leakage current compensation (DLCC) [16] using a Keysight B1500A semiconductor device analyzer. The maximum values of the triangular pulse sequence are V_{PGM} and V_{ERS} , respectively. The P_F — V_G curve [see Fig. 3(i)] shows single loop hysteresis with consistent two peaks in the switching current versus gate voltage (I_{sw} — V_G) curves—typical characteristics of an FE material [see Fig. 3(i)]. The MW obtained from the P_F — V_G loop is denoted by ΔV_P here. ΔP_F , the switched polarization charge of the FE layer, can also be determined from the P_F — V_G loops as indicated in Fig. 3(i).

ΔV_t obtained from step 1 and ΔP_F from step 2 which are plotted as functions of V_{ERS} for different values of V_{PRG} are shown in Fig. 4(a) and (b), respectively. It can be observed from these figures that at a given V_{PRG} , an increase in $|V_{\text{ERS}}|$ leads to a monotonic increase in ΔP_F , while ΔV_t increases until $|V_{\text{ERS}}| \approx 6.5 \text{ V}$, beyond which it starts to decrease. This indicates that while $|V_{\text{ERS}}|$ above 6.5 V results in further switching of polarization, the increased trapping effects at this voltage range overcompensate the effects of the increased ferroelectric polarization on the semiconductor charge. As a result, the ΔV_t — ΔP_F curves for different values of V_{PRG} as shown in Fig. 4(c) have dome shapes, with the peak of the dome increasing with the increase in V_{PRG} . The slope of ΔV_t — ΔP_F curves is the switched polarization to memory window conversion efficiency η_{MW} , as defined in Section I. Fig. 5(a) shows the phase plot of η_{MW} in the (V_{ERS} , V_{PGM}) plane. The ON-state capacitance (C_{ON}) of the FEFET was measured to be $1.34 \mu\text{F}/\text{cm}^2$ [see Fig. 4(d)] using the impedance analyzer (E4990A). The switched polarization to semiconductor charge conversion efficiency $\eta_C = C_{\text{ON}}\eta_{\text{MW}}$

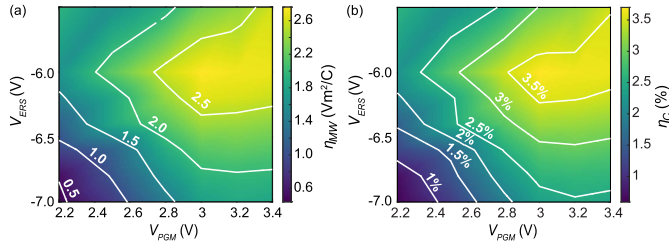


Fig. 5. Phase maps of two efficiency parameters introduced in Section I. (a) $\eta_{MW} = \Delta V_i / \Delta P_F$ and (b) $\eta_C = \Delta Q_S / \Delta P_F$ for different V_{PGM} and V_{ERS} combinations.

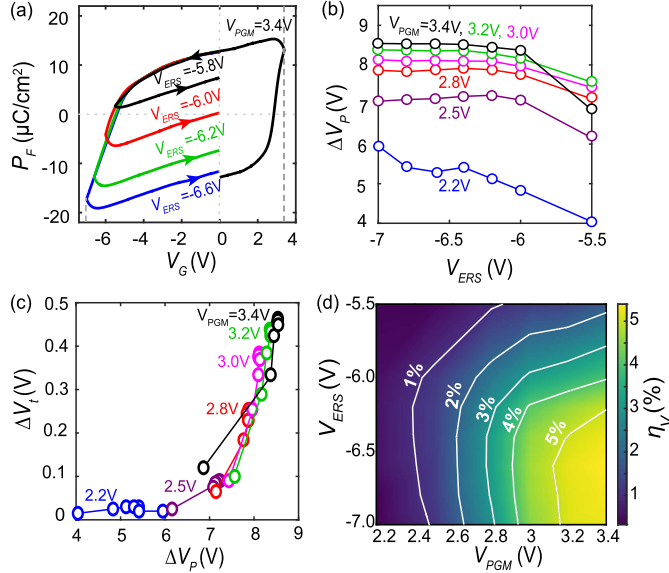


Fig. 6. (a) P_F - V_G hysteresis loops including minor loops of the FEFET. The dashed gray lines show V_{PGM} (≈ 3.4 V) and V_{ERS} (≈ -6.6 V) at which the P_F - V_G goes to the major loop. (b) ΔV_P at different V_{PGM} and V_{ERS} combinations obtained from the P_F - V_G hysteresis loops. (c) Variation of ΔV_i versus ΔV_P for different values of V_{PGM} in ZrO_2 FET. (d) Phase map of memory ratio $\eta_V = \Delta V_i / \Delta V_P$ for different V_{PGM} and V_{ERS} combinations.

is shown as a phase plot in the (V_{ERS}, V_{PGM}) plane in Fig. 5(b). We observe in Fig. 5(b) that in this particular device, only 1%–3.5% of the switched ferroelectric polarization gets reflected as the semiconductor charge.

Fig. 6(a) Major hysteresis loop in the P_F - V_G characteristics obtained by sweeping V_G between 3.4 and -6.6 V as well as a number of minor loops obtained by sweeping V_G in a smaller voltage range. It is clearly observed that the width of the major hysteresis loop ΔV_P (≈ 9 V) is much higher than the memory window ΔV_i (≈ 0.7 V) obtained for the corresponding write voltages $V_{PGM} = 3.4$ V and $V_{ERS} = -6.6$ V. Fig. 6(b) plots ΔV_P as a function of V_{ERS} for different values of V_{PGM} , while Fig. 6(c) plots ΔV_i as a function of ΔV_P . Finally, the voltage conversion efficiency, $\eta_V = \Delta V_i / \Delta V_P$, is plotted in Fig. 6(d). It is observed that η_V is only 2%–5% of its theoretical limit (≈ 1). The measurements were performed without waking up the devices, since the values of ΔP_F and ΔV_P and hence the efficiency remain approximately the same before and after wakeup.

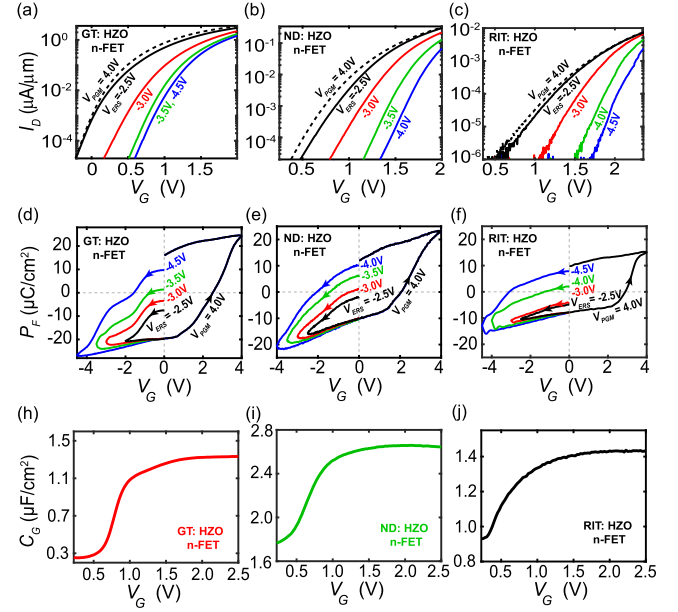


Fig. 7. I_D - V_G curves for $Hf_{0.5}Zr_{0.5}O_2$ n-FEFETs fabricated at (a) GT, (b) UND, and (c) RIT. P_F - V_G hysteresis loops for Hf0.5Zr0.5O2 n-FEFETs fabricated at (d) GATech, (e) UND, and (f) RIT. C_G - V_G curves for the same n-FEFETs fabricated at (g) GATech, (h) UND, and (i) RIT.

IV. COMPARISON OF EFFICIENCY BETWEEN DIFFERENT FEFETs

With the protocol for the measurement of FEFET efficiency metrics established in the previous section, we now compare different FEFETs of different types and fabricated in different facilities in terms of efficiencies. In addition to the p-type ZrO_2 -based FEFET discussed in the previous section, we consider an n-type FEFET with a 10-nm $Hf_{0.5}Zr_{0.5}O_2$ ferroelectric layer fabricated at the Georgia Tech (GT) cleanroom, an n-type FEFET with a 10-nm $Hf_{0.5}Zr_{0.5}O_2$ ferroelectric layer fabricated at the University of Notre Dame (UND) cleanroom, and an n-type FEFET with a 10-nm $Hf_{0.5}Zr_{0.5}O_2$ ferroelectric layer fabricated at the Rochester Institute of Technology (RIT) cleanroom. Fig. 7 shows the I_D - V_G , P_F - V_G , and C_G - V_G characteristics of these devices. Fig. 8(a) and (b) shows the comparison of $\Delta V_i - \Delta P_F$ and $\Delta V_i - \Delta V_P$ characteristics among these devices. The polarization to MW conversion ratio (η_{MW}) for all the FETs [see Fig. 8(c)] stays below 5.5 Vm^2/C . The voltage conversion efficiency η_V stays below 20%, and the charge conversion efficiency η_C stays below 10% for the four FEFETs as evident in Fig. 8(d) and (e), respectively. The maximum values for η_{MW} , η_V , and η_C for the four devices are shown in Fig. 8(f)–(h), respectively. The $(\eta_{MW})_{max}$ ranges from 2.5 to 5.5 Vm^2/C [see Fig. 8(f)] which indicates that if the remnant polarization is 20 $\mu C/cm^2$, the maximum MW will be only 0.5–1.1 V. Fig. 8(g) also shows that the n-type HZO FEFETs from all three facilities (GT, ND, and RIT) are observed to be more efficient with regard to η_V than the p-type ZrO_2 FEFET, which is mostly due to the much higher voltage required to switch the ferroelectric ZrO_2 . Finally, it is evident from Fig. 8(h) that maximum 4%–10% of the switched FE polarization is translated to the semiconductor as inversion carriers.

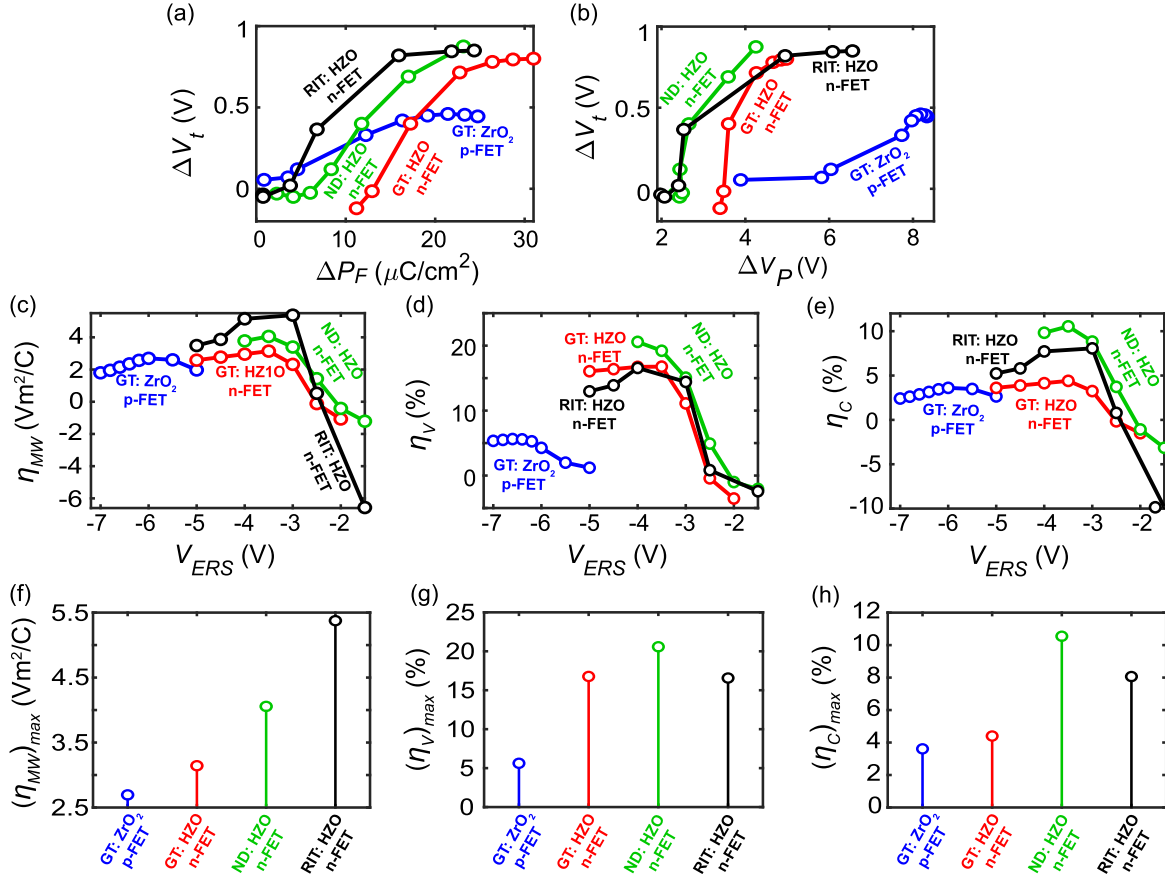


Fig. 8. (a) ΔV_t versus ΔP_F , and (b) ΔV_t versus ΔV_P for four different FEFETs (blue: ZrO_2 p-FEFET and red: $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ n-FEFET fabricated at GT, green: $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ n-FEFET fabricated at UND, and black: $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ n-FEFET fabricated at RIT). (c) $\eta_{MW} = \Delta V_t / \Delta P_F$, (d) $\eta_V = \Delta V_t / \Delta V_P$, and (e) $\eta_C = \Delta Q_S / \Delta P_F$ versus V_{ERS} for the four FEFETs. (f) $\eta_{MW, \max}$ (range ~ 2.5 – $5.5 \text{ Vm}^2/\text{C}$), (g) $\eta_{V, \max}$ (range ~ 4 – 10%), and (h) $\eta_{C, \max}$ (range ~ 4 – 10%) for the four devices. These plots were evaluated for the four FEFETs at the minimum V_{PGM} which gives the maximum ΔV_t : $V_{PGM} = 3.4 \text{ V}$ for the ZrO_2 p-FEFET and $V_{PGM} = 4 \text{ V}$ for the $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ n-FEFETs.

We note that the P_F – V_G curves shown in Figs. 3(i) and 7(d)–(f) do not have an odd symmetry [i.e., $P_F(V_G) \neq -P_F(-V_G)$], as one would expect for the case of an ideal metal–ferroelectric–metal (MFM) capacitor with top and bottom electrodes with the same work-functions and no trapped charges. To that end, we note that such a symmetry is not guaranteed in the P_F – V_G characteristics of a metal–ferroelectric–insulator–semiconductor (MFIS) structure, i.e., the gate-stack of an FEFET—even in the ideal case. This is because the semiconductor charge–voltage characteristics of a standard metal–oxide–semiconductor (MOS) structure includes accumulation, depletion, and inversion regions and hence is not odd-symmetric. The presence of bulk and interface traps and difference in the work-function between the metal and the semiconductor layer may add further to the asymmetry. All these factors lead to a situation that the coercive voltages in the FEFET P – V_G curves are not symmetric with respect to $V_G = 0$ (i.e., $V_{C+} \neq |V_{C-}|$). Such an asymmetry is particularly more pronounced in the FEFET P_F – V_G characteristics shown in Figs. 3(i) and 7(f), wherein $|V_{C-}| > V_{C+}$. In both of these cases, V_G had to be swept to a larger value in the negative side than in the positive side. As such, P_F at the most negative V_G is larger in magnitude than that at the most positive V_G .

V. CONCLUSION

In summary, we established a generalized experimental framework to quantify the efficiency of practical FEFETs in converting the switched ferroelectric polarization into the memory window. This was achieved by a measurement protocol where the memory window, the switched polarization, and the width of the polarization versus gate voltage hysteresis characteristics were all measured on the same devices, subsequently for a given value of program and erase voltages. Based thereon, we calculated three efficiency metrics: 1) switched ferroelectric polarization ΔP_F to memory window conversion ratio $\eta_{MW} = \Delta V_t / \Delta P_F$; 2) switched ferroelectric polarization to the semiconductor charge (ΔQ_S) conversion efficiency or the charge conversion efficiency $\eta_C = \Delta Q_S / \Delta P_F$; and 3) the ratio of memory window to the width of polarization versus gate voltage hysteresis characteristics (ΔV_P) or the voltage conversion efficiency $\eta_V = \Delta V_t / \Delta V_P$. We measure and compare these efficiency metrics in n-type and p-type FEFETs, fabricated in different facilities in different groups. In all the cases, we find that the maximum values of polarization to MW conversion ratio $(\eta_{MW})_{\max}$, charge conversion efficiency $(\eta_C)_{\max}$, and voltage conversion efficiency $(\eta_V)_{\max}$ are in the range: 2.5 – $5.5 \text{ Vm}^2/\text{C}$, 4% – 10% , and 5% – 20% , respectively. Our work is a step toward establishing efficiency metrics

for ferroelectric devices which will allow for the community to benchmark FEFET results generated all across the globe, leading toward an informed and steady progress of the field.

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